SOFTWARE REQUIREMENTS SPECIFICATION

LOCUS

Living Space and Occupancy Control Unified System

Enterprise Campus Housing Platform • Room & Asset Management

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, external-interface, data, and quality requirements for LOCUS (Living Space and Occupancy Control Unified System), an enterprise campus housing platform for room occupancy, student allotment, room-level asset custody, check-in verification, and No-Dues clearance. It is intended to provide a common, testable baseline for stakeholders, designers, developers, and testers.

1.2 Scope

LOCUS will digitize hostel room hierarchy and occupancy, student room allotment, mutually consented room swaps, furniture inventory, QR/barcode-based audits, mandatory move-in inventory review, and exit clearance. The system will replace paper registers, verbal allotments, and untracked room transitions with a digital record of occupancy and asset accountability. Routine civil and electrical maintenance workflows are explicitly outside LOCUS scope and remain within the campus Computerized Maintenance Management System (CMMS).

1.3 Intended Audience

The intended audience includes Students, Caretakers, Wardens, the Central Estate Office, institutional IT/identity administrators, project stakeholders, software developers, quality-assurance personnel, and maintainers.

1.4 Document Conventions

The words “shall” and “must” indicate mandatory requirements. Each requirement has a unique identifier. Requirements that cannot be fully determined from the supplied project overview are identified as items requiring stakeholder confirmation; they are not silently treated as settled design decisions.

1.5 Definitions and Abbreviations

LOCUS — Living Space and Occupancy Control Unified System; SRS — Software Requirements Specification; PWA — Progressive Web App; RBAC — Role-Based Access Control; SSO — Single Sign-On; LDAP — Lightweight Directory Access Protocol; CMMS — Computerized Maintenance Management System; No-Dues — institutional clearance certificate required at exit/graduation; QR — Quick Response code.

1.6 References

Primary source: “LOCUS Project Overview — Enterprise Campus Housing Platform • Room & Asset Management,” supplied project idea PDF. This document is organized using a conventional IEEE-style SRS structure; it is not a claim of formal certification against a specific IEEE standard edition.

2. Overall Description

2.1 Product Perspective

LOCUS is a multi-tier responsive web application with a mobile Progressive Web App. It integrates with institutional SSO through Google Workspace and/or LDAP, applies RBAC to four user groups, and stores records in a relational data tier. The organizational hierarchy is Hostel → Block → Floor → Room. Room records are connected to an inventory ledger that tracks individual physical assets and their movements.

2.2 Product Functions

The system shall: (a) manage hostel/block/floor/room records and room states; (b) enforce bed limits; (c) support batch allotment using student preferences and configured seniority/category rules; (d) manage room swaps with consent from both students and Warden authorization; (e) associate baseline furniture packages and individual assets with rooms; (f) support QR/barcode room audits and movement logging; (g) enforce a 48-hour move-in inventory review; and (h) hold No-Dues clearance until asset verification is complete and discrepancies are resolved.

2.3 User Classes and Characteristics

Student: accesses relevant allotment/room information, provides consent for a proposed mutual swap, and participates in move-in inventory review. Caretaker: performs permitted room and asset verification and records audit findings. Warden: authorizes room swaps and performs authorized oversight. Central Estate Office: administers central residential records, room hierarchy, allotment operations, and occupancy/asset governance. The institution must approve the detailed role-permission matrix before implementation.

2.4 Operating Environment

The application shall be accessible through a responsive web interface and a mobile PWA. QR/barcode audits require a compatible device camera and permission to use it. Authentication depends on an institutional Google Workspace and/or LDAP identity service. Specific browser versions, mobile operating systems, hosting platform, and network requirements are not stated in the source and require confirmation.

2.5 Design and Implementation Constraints

The system shall use a multi-tier architecture and relational data model; enforce the hierarchy Hostel → Block → Floor → Room; prevent occupancy beyond configured bed limits; preserve the inventory ledger as an authoritative record; and keep routine civil/electrical maintenance workflows outside LOCUS.

2.6 User Documentation

The supplied overview does not specify user manuals or training materials. Before deployment, the project owner should decide whether to provide role-specific quick-start guides for Students, Caretakers, Wardens, and Central Estate Office staff.

2.7 Assumptions and Dependencies

The institution will provide and maintain the configured SSO identity service. Authorized institutional staff will define room capacities, allotment rules, baseline furniture packages, and clearance procedures. Users conducting field audits will have compatible camera-equipped devices and grant camera access when prompted.

3. System Features and Functional Requirements

3.1 Authentication and Role-Based Access Control

FR-01: The system shall authenticate users through the institutional SSO integration configured for Google Workspace and/or LDAP.\nFR-02: The system shall support the user groups Student, Caretaker, Warden, and Central Estate Office.\nFR-03: The system shall authorize each protected operation according to the user's assigned role and approved permission matrix.\nFR-04: The system shall prevent a user from performing an operation for which that user's role has not been authorized.

3.2 Room Hierarchy and Occupancy Tracking

FR-05: The system shall represent residential locations using the hierarchy Hostel → Block → Floor → Room.\nFR-06: The system shall store a configured bed/occupancy limit for each room.\nFR-07: The system shall maintain one of the defined room states: Vacant, Partially Occupied, Fully Occupied, Reserved, or Decommissioned.\nFR-08: The system shall reject an allotment or occupancy change that would cause the number of occupants to exceed the room's configured bed limit.\nFR-09: The system shall update the room's occupancy record when a valid allotment, transfer, check-in, or check-out changes its occupants.\nFR-10: The system shall not make a Decommissioned room available for a new allotment. The precise process for reserving or decommissioning rooms requires stakeholder confirmation.

3.3 Student Allotment and Mutual Room Transfers

FR-11: The system shall support batch allotment of students using recorded preferences and institution-configured seniority and category rules.\nFR-12: The system shall apply the configured allotment rules consistently to all records processed in a batch. Exact rule precedence and tie-breaking logic must be supplied by the institution before implementation.\nFR-13: The system shall support a mutual room-swap request between two students.\nFR-14: The system shall record consent from both students before a mutual room swap can proceed to authorization.\nFR-15: The system shall require Warden authorization before finalizing a room swap.\nFR-16: The system shall not change

room occupancy records when either student has not consented or Warden authorization has not been granted.\nFR-17: After an authorized swap, the system shall update the affected student-room associations and occupancy records as one logically consistent operation.

3.4 Room Asset Inventory and QR/Barcode Custody

FR-18: The system shall associate a standardized baseline furniture package with each applicable room. The source identifies beds, desks, chairs, and wardrobes as example asset types.\nFR-19: The system shall maintain an individual record for each tracked physical asset and associate it with its current room.\nFR-20: The system shall support QR/barcode tags for room-level asset audits.\nFR-21: On a permitted user's successful scan of a room-door QR/barcode, the mobile PWA shall retrieve the corresponding room's verified asset checklist.\nFR-22: The system shall record inter-room movements of tracked assets in the inventory ledger.\nFR-23: The system shall preserve historical asset-movement records against silent overwrite or deletion during ordinary use. The authorized correction process and audit fields must be defined during detailed design.\nFR-24: The system shall allow an authorized user to record an audit result and identify asset discrepancies. Specific discrepancy categories and evidence requirements require stakeholder confirmation.

3.5 Check-in Inventory Verification

FR-25: The system shall require a move-in inventory review to be completed within 48 hours of the student's check-in.\nFR-26: The system shall record the review status and the outcome of the room asset verification.\nFR-27: Where a discrepancy is recorded during the review, the system shall retain the discrepancy against the relevant room/asset record until it is resolved or formally dispositioned under an institution-approved process.

3.6 No-Dues / Exit Clearance

FR-28: The system shall prevent issuance of an institutional No-Dues clearance certificate until the required room assets have been physically verified.\nFR-29: If an asset shortage or unresolved discrepancy exists, the system shall place the clearance on hold.\nFR-30: The system shall allow clearance to proceed only after the required verification is complete and the hold-causing discrepancy has been resolved.\nFR-31: The system shall retain the clearance status and its relationship to the relevant student, room, and asset verification records.

3.7 Record Integrity and Auditability

FR-32: The system shall maintain an authoritative digital record of room occupancy and room-level asset custody.\nFR-33: The system shall preserve consistency between room capacity, allotment, occupancy, transfer, and asset records.\nFR-34: The system shall prevent an operation from creating an occupancy count above the configured room limit, including when multiple allotments are processed concurrently.\nFR-35: The system shall make relevant verification and clearance outcomes available to authorized users. Detailed reports, export formats, retention periods, and audit-log fields require confirmation.

4. External Interface Requirements

4.1 User Interface

UI-01: The system shall provide a responsive web interface.\nUI-02: The system shall provide a mobile PWA suitable for room and asset audits.\nUI-03: The interface shall present available operations according to the authenticated user's role.\nUI-04: The interface shall clearly indicate room state, occupancy against bed limit, swap approval status, asset verification status, and No-Dues hold status. Exact layouts, branding, accessibility criteria, and supported languages require stakeholder approval.

4.2 Hardware Interface

HW-01: For QR/barcode audits, the mobile PWA shall request access to the device camera and use it to scan supported room/asset codes when permission and compatible hardware are available.

4.3 Software Interface

SW-01: The system shall integrate with the institution's configured Google Workspace and/or LDAP SSO service.\nSW-02: The system shall persist room, student-room association, asset, movement, verification, and

clearance records in a relational data tier.\nSW-03: No integration with the campus CMMS is required for civil/electrical maintenance workflows by this specification; those workflows remain outside LOCUS. Integration protocols and any other campus-system interfaces are not specified in the source.

4.4 Communications Interface

The application will be accessed over web technologies. Specific network protocols, API contracts, encryption configuration, and offline synchronization behavior are not specified by the project overview and must be selected and documented during design.

5. Non-Functional Requirements

5.1 Performance and Capacity

NFR-01: During a simulated batch-allotment stress test, the system shall sustain 3,500 concurrent sessions while maintaining page response times below 1.5 seconds.\nNFR-02: During field conditions, scanning a room-door QR code shall retrieve the verified room asset checklist in under 200 milliseconds.\nNFR-03: The test plan shall define the measurement boundary, hardware/software environment, workload, and percentile/measurement method used to verify NFR-01 and NFR-02.

5.2 Integrity and Reliability

NFR-04: In the defined capacity-lock test, the system shall reject every attempted over-allocation beyond a room's configured bed limit; the target is 100% boundary integrity.\nNFR-05: In the defined No-Dues test, a simulated asset shortage shall cause clearance to remain on hold until the discrepancy is resolved.\nNFR-06: The system shall preserve consistent room occupancy and asset associations across allotment, transfer, audit, and clearance operations.

5.3 Security and Authorization

NFR-07: The system shall use institutional SSO and RBAC for the four user groups defined in this SRS.\nNFR-08: Access to student, room, asset, and clearance records shall be restricted to authenticated users with the required permissions.\nNFR-09: Encryption at rest/in transit, session lifetime, security monitoring, backup controls, and detailed audit logging must be specified and approved by institutional IT before production deployment; the source overview does not provide values for these controls.

5.4 Usability and Compatibility

NFR-10: The web interface shall adapt to supported desktop and mobile viewport sizes.\nNFR-11: The mobile audit workflow shall support QR/barcode scanning on compatible camera-equipped devices.\nNFR-12: Supported browsers, mobile operating systems, accessibility conformance target, and offline behavior must be agreed and documented before acceptance testing.

5.5 Architecture and Maintainability

NFR-13: The system shall follow a multi-tier architecture and use a relational data model.\nNFR-14: Routine civil and electrical maintenance features shall remain outside the LOCUS product boundary and be handled by the campus CMMS.\nNFR-15: The implementation shall keep business rules for capacity limits, allotment, swap approval, inventory verification, and No-Dues clearance testable as separate requirements.

6. Data Requirements

6.1 Core Data Entities

The following logical entities are implied by the project overview and functional requirements: Hostel, Block, Floor, Room, Student, User/Role, Allotment, Room Swap/Transfer, Asset Type, Physical Asset, QR/Barcode Tag, Asset Movement, Check-in Verification, Discrepancy, and No-Dues Clearance. This is a logical inventory of information, not a finalized physical database schema.

6.2 Data Relationships

Each Room shall belong to one Floor; each Floor shall belong to one Block; each Block shall belong to one Hostel. Each tracked asset shall have a current room association, with movement history recorded when it changes rooms. Allotments and swaps shall associate students with rooms over the relevant occupancy period. Verification and clearance records shall refer to the relevant student, room, and asset records.

6.3 Data Integrity Rules

The system shall enforce room bed limits and maintain consistent occupancy counts. A room swap shall not be committed until both student consents and Warden authorization are recorded. An asset movement shall update the current asset-room association while retaining the movement history. No-Dues clearance shall not be issued while mandatory physical verification is incomplete or an applicable discrepancy remains unresolved.

6.4 Data Items Requiring Definition

The project overview does not define mandatory field lists, identifier formats, data retention periods, archival/deletion rules, student-data privacy classifications, asset condition codes, discrepancy severity, or authorized correction procedures. These must be finalized by the project owner and institutional IT.

7. Use Cases

UC-01 Batch Room Allotment

Primary actor: authorized allotment administrator (role to be confirmed within the Central Estate Office). Preconditions: room capacities are configured; student preferences and approved seniority/category rules are available. Main flow: administrator starts a batch; system applies configured rules; system assigns students without exceeding room limits; system records allotments and updates room occupancy. Exception: any assignment that would exceed a room limit is rejected and the affected record is not committed.

UC-02 Mutual Room Swap

Primary actors: two students and a Warden. Preconditions: the students and rooms are eligible under institution-defined rules. Main flow: a swap is requested; the system captures consent from both students; the request is routed for Warden authorization; if approved, the system updates both room associations and occupancy records. Exceptions: if either consent is missing or authorization is denied, the swap is not finalized.

UC-03 Room Asset Audit

Primary actor: authorized Caretaker or other permitted user. Preconditions: the room QR/barcode and its asset checklist exist; the user has audit permission. Main flow: user scans the room-door code; system retrieves the checklist; user physically checks assets and records findings; the system records discrepancies and retains the audit outcome.

UC-04 Check-in Inventory Review

Primary actor: Student and/or authorized Caretaker, subject to the institution-approved workflow. Preconditions: the student has checked into a room. Main flow: the required inventory review is completed within 48 hours; the result is recorded; any discrepancy is linked to the relevant room/asset record.

UC-05 No-Dues Clearance

Primary actor: authorized clearance officer (role to be confirmed). Preconditions: an exiting student has a room/asset record. Main flow: required assets are physically verified; discrepancies are checked; clearance is permitted after all required conditions are met. Exception: if an asset shortage or unresolved discrepancy exists, the system holds the certificate.

8. Verification and Acceptance Criteria

AT-01 Concurrency and Load

Method: run a simulated stress test during batch allotment. Pass criterion: sustain 3,500 concurrent sessions with page response below 1.5 seconds, under the test conditions documented in the test plan.

AT-02 Capacity Lock

Method: attempt to allocate occupants beyond a room's configured bed limit, including concurrent requests. Pass criterion: all over-limit attempts are rejected and no committed state exceeds the configured capacity.

AT-03 Asset QR Audits

Method: scan room-door QR codes using the mobile camera in field conditions. Pass criterion: the verified room asset checklist is fetched in under 200 milliseconds.

AT-04 No-Dues Clearance Gate

Method: run an end-to-end checkout/clearance test with a simulated asset shortage. Pass criterion: the exit certificate remains on hold until the discrepancy is resolved.

AT-05 Mutual Swap Authorization

Method: attempt a swap with missing consent and then with both consents but without Warden authorization. Pass criterion: neither attempt changes occupancy; the swap proceeds only after both consents and Warden authorization.

AT-06 Check-in Review Deadline

Method: create check-in records and verify the review workflow at and around the 48-hour deadline. Pass criterion: the system records whether the mandatory inventory review was completed within the required 48-hour window. The handling of overdue reviews must follow a stakeholder-approved policy.

9. Open Issues Requiring Stakeholder Approval

OI-01 Role Permissions

Approve the complete permission matrix for Student, Caretaker, Warden, and Central Estate Office.

OI-02 Allotment Rules

Specify exact preference handling, seniority/category precedence, tie-breaking, batch failure handling, and whether allotments can be manually overridden.

OI-03 Room State Rules

Define the transition rules for Vacant, Partially Occupied, Fully Occupied, Reserved, and Decommissioned states.

OI-04 Asset and Discrepancy Policy

Define asset identifiers, condition categories, discrepancy severity, evidence requirements, resolution authority, and correction procedures.

OI-05 No-Dues Authority

Confirm the role permitted to issue clearance and the exact evidence required to release a hold.

OI-06 Security and Operations

Define encryption, session timeout, audit-log policy, backups, recovery objectives, hosting, monitoring, and data retention.

OI-07 Compatibility and Interfaces

Confirm supported browsers/devices, accessibility and language targets, offline PWA behavior, API contracts, and any required integrations beyond SSO.

OI-08 Performance Test Definition

Define workload profile, test environment, measurement boundary, and method for verifying the stated performance targets.

OI-09 Project Team Details

The supplied one-page LOCUS Project Overview does not contain team member names or SIDs. Populate these fields from the project proposal PDF before submission; no names or SIDs have been invented.

10. Appendix

A. Requirement Traceability Summary

Room hierarchy and room states: FR-05 to FR-10. Student allotment and swaps: FR-11 to FR-17. Room assets and QR custody: FR-18 to FR-24. Check-in review: FR-25 to FR-27. No-Dues clearance: FR-28 to FR-31. Record integrity: FR-32 to FR-35. Performance and verification targets: NFR-01 to NFR-06 and AT-01 to AT-06.

B. Project Team / Submission Details

Project name: LOCUS — Living Space and Occupancy Control Unified System. Course / institution: [Insert from project proposal if applicable]. Team members and SIDs: [Copy exact names and SIDs from the project proposal PDF]. Prepared from: LOCUS Project Overview supplied by the project owner. Version: 1.0. Status: Draft for stakeholder review.